

# Solving Recurrence Relations Using Generating Functions

**Method overview:** For a recurrence relation in  $a_r$ , define  $G(x) = \sum_{r=0}^{\infty} a_r x^r$ . Multiply the recurrence by  $x^r$ , sum over the valid range, rewrite each sum in terms of  $G(x)$ , solve algebraically for  $G(x)$ , decompose into partial fractions, then read off  $a_r$  as the coefficient of  $x^r$  using standard series expansions:

$$\frac{1}{1 - cx} = \sum_{r=0}^{\infty} c^r x^r, \quad \frac{1}{(1 - x)^2} = \sum_{r=0}^{\infty} (r + 1) x^r, \quad \frac{1}{(1 - x)^3} = \sum_{r=0}^{\infty} \binom{r + 2}{2} x^r$$

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## Problem 1

**Solve:**  $a_r - 3a_{r-1} + 2a_{r-2} = 0$ , for  $r \geq 2$ , with  $a_0 = 2, a_1 = 3$ .

**Step 1 — Set up.** Let  $G(x) = \sum_{r=0}^{\infty} a_r x^r$ . Multiply by  $x^r$  and sum from  $r = 2$ :

$$\sum_{r=2}^{\infty} a_r x^r - 3 \sum_{r=2}^{\infty} a_{r-1} x^r + 2 \sum_{r=2}^{\infty} a_{r-2} x^r = 0$$

**Step 2 — Rewrite in terms of  $G(x)$ .**

$$\sum_{r=2}^{\infty} a_r x^r = G(x) - 2 - 3x, \quad \sum_{r=2}^{\infty} a_{r-1} x^r = x(G(x) - 2), \quad \sum_{r=2}^{\infty} a_{r-2} x^r = x^2 G(x)$$

$$(G(x) - 2 - 3x) - 3x(G(x) - 2) + 2x^2 G(x) = 0$$

**Step 3 — Solve for  $G(x)$ .**

$$G(x)(1-x)(1-2x) = 2-3x \quad \implies \quad G(x) = \frac{2-3x}{(1-x)(1-2x)}$$

**Step 4 — Partial fractions.**  $\frac{2-3x}{(1-x)(1-2x)} = \frac{A}{1-x} + \frac{B}{1-2x}$

- $x = 1: A = 1$
- $x = \frac{1}{2}: B = 1$

$$G(x) = \frac{1}{1-x} + \frac{1}{1-2x}$$

**Step 5 — Extract coefficients.**

$$a_r = 1^r + 2^r$$

**Verify:**  $a_0 = 1 + 1 = 2 \checkmark$      $a_1 = 1 + 2 = 3 \checkmark$

$$\boxed{a_r = 1 + 2^r}$$

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## Problem 2

**Solve:**  $a_r - 2a_{r-1} + a_{r-2} = 2^r$ , for  $r \geq 2$ , with  $a_0 = 2, a_1 = 1$ .

**Step 1 — Set up.**

$$\sum_{r=2}^{\infty} a_r x^r - 2 \sum_{r=2}^{\infty} a_{r-1} x^r + \sum_{r=2}^{\infty} a_{r-2} x^r = \sum_{r=2}^{\infty} 2^r x^r$$

**Step 2 — Right side.**

$$\sum_{r=2}^{\infty} (2x)^r = \frac{1}{1-2x} - 1 - 2x = \frac{4x^2}{1-2x}$$

**Left side:**

$$(G(x) - 2 - x) - 2x(G(x) - 2) + x^2G(x) = \frac{4x^2}{1-2x}$$

**Step 3 — Solve for  $G(x)$ .**

$$G(x)(1-x)^2 = \frac{(2-3x)(1-2x) + 4x^2}{1-2x} = \frac{2-7x+10x^2}{1-2x}$$

$$G(x) = \frac{10x^2 - 7x + 2}{(1-x)^2(1-2x)}$$

**Step 4 — Partial fractions.**  $\frac{10x^2 - 7x + 2}{(1-x)^2(1-2x)} = \frac{A}{1-x} + \frac{B}{(1-x)^2} + \frac{C}{1-2x}$

- $x = 1: B = -5$

- $x = \frac{1}{2}: C = 4$
- $x = 0: A = 3$

$$G(x) = \frac{3}{1-x} - \frac{5}{(1-x)^2} + \frac{4}{1-2x}$$

**Step 5 — Extract coefficients.**

$$a_r = 3(1)^r - 5(r+1)(1)^r + 4(2^r) = 2^{r+2} - 5r - 2$$

**Verify:**  $a_0 = 4 - 0 - 2 = 2 \checkmark$      $a_1 = 8 - 5 - 2 = 1 \checkmark$

$$\boxed{a_r = 2^{r+2} - 5r - 2}$$

### Problem 3

**Solve:**  $a_r - 5a_{r-1} + 6a_{r-2} = 2$ , for  $r \geq 2$ , with  $a_0 = 1, a_1 = 1$ .

**Step 1 — Set up.**

$$\sum_{r=2}^{\infty} a_r x^r - 5 \sum_{r=2}^{\infty} a_{r-1} x^r + 6 \sum_{r=2}^{\infty} a_{r-2} x^r = \sum_{r=2}^{\infty} 2x^r$$

**Step 2 — Right side.**

$$2 \sum_{r=2}^{\infty} x^r = \frac{2x^2}{1-x}$$

**Left side:**

$$(G(x) - 1 - x) - 5x(G(x) - 1) + 6x^2 G(x) = \frac{2x^2}{1-x}$$

**Step 3 — Solve for  $G(x)$ .**

$$G(x)(1 - 2x)(1 - 3x) - 1 + 4x = \frac{2x^2}{1-x}$$

$$G(x)(1-2x)(1-3x) = \frac{(1-4x)(1-x) + 2x^2}{1-x} = \frac{1-5x+6x^2}{1-x} = \frac{(1-2x)(1-3x)}{1-x}$$

**Clean cancellation:**

$$G(x) = \frac{1}{1-x}$$

**Step 4 — Extract coefficients.** The coefficient of  $x^r$  in  $\frac{1}{1-x}$  is simply 1.

**Verify:**  $1 - 5(1) + 6(1) = 2 \checkmark$  (holds for every  $r$ );  $a_0 = a_1 = 1 \checkmark$

$a_r = 1$

## Problem 4

**Solve:**  $a_r - 8a_{r-1} = 10^{r-1}$ , with  $a_1 = 9$ .

**Step 1 — Set up.** Since only  $a_1$  is given, let  $A(x) = \sum_{r=1}^{\infty} a_r x^r$ . Multiply by  $x^r$  and sum from  $r = 2$ :

$$\sum_{r=2}^{\infty} a_r x^r - 8 \sum_{r=2}^{\infty} a_{r-1} x^r = \sum_{r=2}^{\infty} 10^{r-1} x^r$$

**Step 2 — Rewrite.**

$$\sum_{r=2}^{\infty} a_r x^r = A(x) - 9x, \quad \sum_{r=2}^{\infty} a_{r-1} x^r = xA(x), \quad \sum_{r=2}^{\infty} 10^{r-1} x^r = \frac{10x^2}{1-10x}$$

$$(A(x) - 9x) - 8xA(x) = \frac{10x^2}{1-10x}$$

**Step 3 — Solve for  $A(x)$ .**

$$A(x)(1-8x) = 9x + \frac{10x^2}{1-10x} = \frac{9x-80x^2}{1-10x}$$



$$A(x) = \frac{9x - 80x^2}{(1 - 8x)(1 - 10x)} = x \left[ \frac{9 - 80x}{(1 - 8x)(1 - 10x)} \right]$$

**Step 4 — Partial fractions.**  $\frac{9 - 80x}{(1 - 8x)(1 - 10x)} = \frac{C}{1 - 8x} + \frac{D}{1 - 10x}$

- $x = \frac{1}{10}: D = 5$
- $x = \frac{1}{8}: C = 4$

$$A(x) = \frac{4x}{1 - 8x} + \frac{5x}{1 - 10x}$$

**Step 5 — Extract coefficients.**

$$a_r = 4(8^{r-1}) + 5(10^{r-1}) = \frac{8^r}{2} + \frac{10^r}{2}$$

**Verify:**  $a_1 = 4 + 5 = 9 \checkmark$ ; recurrence at  $r = 2$ :  $a_2 = 8(9) + 10 = 82$ ; formula gives  $(64 + 100)/2 = 82 \checkmark$

$$a_r = \frac{8^r + 10^r}{2}$$

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### Problem 5

**Solve:**  $a_n = 3a_{n-1} + 2n$ , with  $a_0 = 1$ .

**Step 1 — Set up.**

$$\sum_{n=1}^{\infty} a_n x^n = 3 \sum_{n=1}^{\infty} a_{n-1} x^n + 2 \sum_{n=1}^{\infty} n x^n$$

Using  $\sum_{n=1}^{\infty} n x^n = \frac{x}{(1-x)^2}$ :

$$G(x) - a_0 = 3xG(x) + \frac{2x}{(1-x)^2}$$

**Step 2 — Solve for  $G(x)$ .**

$$G(x)(1 - 3x) = a_0 + \frac{2x}{(1 - x)^2}$$

$$G(x) = \frac{a_0}{1 - 3x} + \frac{2x}{(1 - 3x)(1 - x)^2}$$

**Step 3 — Partial fractions.**  $\frac{2x}{(1 - 3x)(1 - x)^2} = \frac{A}{1 - 3x} + \frac{B}{1 - x} + \frac{D}{(1 - x)^2}$

- $x = 1: D = -1$
- $x = \frac{1}{3}: A = \frac{3}{2}$
- $x = 0: B = -\frac{1}{2}$

$$G(x) = \frac{a_0 + \frac{3}{2}}{1 - 3x} - \frac{1/2}{1 - x} - \frac{1}{(1 - x)^2}$$

**Step 4 — Extract coefficients.**

$$a_n = \left(a_0 + \frac{3}{2}\right) 3^n - n - \frac{3}{2}$$

**Substitute  $a_0 = 1$ :** coefficient of  $3^n$  becomes  $1 + \frac{3}{2} = \frac{5}{2}$

$$a_n = \frac{5}{2} \cdot 3^n - n - \frac{3}{2}$$

**Verify:**  $a_0 = \frac{5}{2} - 0 - \frac{3}{2} = 1 \checkmark$  Recurrence check:  $a_1$  should equal  $3a_0 + 2 = 5$ . Formula:  
 $\frac{5}{2}(3) - 1 - \frac{3}{2} = \frac{15}{2} - \frac{3}{2} - 1 = 6 - 1 = 5 \checkmark$

$$a_n = \frac{5}{2} \cdot 3^n - n - \frac{3}{2}$$


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## Problem 6

**Solve:**  $a_n - a_{n-1} - 2a_{n-2} = n^2$ , with  $a_0 = 1, a_1 = 1$ .

**Step 1 — Set up.**

$$\sum_{n=2}^{\infty} a_n x^n - \sum_{n=2}^{\infty} a_{n-1} x^n - 2 \sum_{n=2}^{\infty} a_{n-2} x^n = \sum_{n=2}^{\infty} n^2 x^n$$

**Step 2 — Right side.** Since  $\sum_{n=0}^{\infty} n^2 x^n = \frac{x(1+x)}{(1-x)^3}$ , subtracting the  $n = 0, 1$  terms (both zero except a linear term):

$$\sum_{n=2}^{\infty} n^2 x^n = \frac{x(1+x)}{(1-x)^3} - x$$

**Left side:**

$$(G(x) - a_0 - a_1 x) - x(G(x) - a_0) - 2x^2 G(x) = \frac{x(1+x)}{(1-x)^3} - x$$

**Step 3 — Solve for  $G(x)$ .** Factor  $1 - x - 2x^2 = (1+x)(1-2x)$ :

$$G(x)(1+x)(1-2x) = a_0 + (a_1 - a_0 - 1)x + \frac{x(1+x)}{(1-x)^3}$$

Dividing through (the  $(1 + x)$  factor cancels neatly in the last term):

$$G(x) = \frac{a_0 + (a_1 - a_0 - 1)x}{(1 + x)(1 - 2x)} + \frac{x}{(1 - x)^3(1 - 2x)}$$

**Step 4 — Partial fractions.** The first term contributes constants  $\frac{C_1}{1 + x} + \frac{C'_2}{1 - 2x}$  (to be pinned down by initial conditions). Decompose the second term:

$$\frac{x}{(1 - x)^3(1 - 2x)} = \frac{A}{1 - 2x} + \frac{B}{1 - x} + \frac{C}{(1 - x)^2} + \frac{D}{(1 - x)^3}$$

- $x = 1: D = -1$
- $x = \frac{1}{2}: A = 4$
- Matching  $x^3$  coefficients:  $B = -2$
- Then  $B + C = -3 \Rightarrow C = -1$

Combining the  $(1 - 2x)$  terms into a single constant  $C_2$ :

$$G(x) = \frac{C_1}{1 + x} + \frac{C_2}{1 - 2x} - \frac{2}{1 - x} - \frac{1}{(1 - x)^2} - \frac{1}{(1 - x)^3}$$

**Step 5 — General coefficient formula** (using  $\frac{1}{(1-x)^3} = \sum \binom{n+2}{2} x^n$ ):

$$a_n = C_1(-1)^n + C_2 \cdot 2^n - 2 - (n+1) - \frac{(n+1)(n+2)}{2} = C_1(-1)^n + C_2 \cdot 2^n - \frac{1}{2}n^2 - \frac{5}{2}n - 4$$

**Step 6 — Solve for  $C_1, C_2$  using  $a_0 = 1, a_1 = 1$ .**

$$\text{At } n = 0: \quad C_1 + C_2 - 4 = 1 \implies C_1 + C_2 = 5$$

$$\text{At } n = 1: \quad -C_1 + 2C_2 - \frac{1}{2} - \frac{5}{2} - 4 = 1 \implies -C_1 + 2C_2 = 8$$

$$\text{Adding: } 3C_2 = 13 \implies C_2 = \frac{13}{3}, \text{ then } C_1 = 5 - \frac{13}{3} = \frac{2}{3}$$

**Verify:**

- $a_0 = \frac{2}{3} + \frac{13}{3} - 4 = \frac{15}{3} - 4 = 1 \checkmark$
- $a_1 = -\frac{2}{3} + \frac{26}{3} - 7 = \frac{24}{3} - 7 = 1 \checkmark$
- Recurrence at  $n = 2$ : expect  $a_2 = a_1 + 2a_0 + 4 = 1 + 2 + 4 = 7$ . Formula:  $\frac{2}{3}(1) + \frac{13}{3}(4) - 2 - 5 - 4 = \frac{54}{3} - 11 = 18 - 11 = 7 \checkmark$

$$a_n = \frac{2}{3}(-1)^n + \frac{13}{3} \cdot 2^n - \frac{1}{2}n^2 - \frac{5}{2}n - 4$$